



Research Paper

Parental input as a mediating pathway for gender differences in early academic skills

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ABSTRACT

The persistent gendered pattern in academic achievement, whereby girls outperform boys in language while boys excel in mathematics, is evident as early as primary school. The literature highlights parental input as a key predictor of child development, yet less is known about its role in the context of gender differences in early academic skills. The present study investigated variability in parental input as a function of child gender. Using longitudinal academic scores from a large sample of Russian first-graders ($N = 1240$, 49.5 % female), as well as their parents' reports on math and language home activities, this study examined parental input as a potential mediator of gender differences in children's language and math skills. Results confirmed the gendered pattern in children's academic skills and revealed that parents were more frequently involved in literacy activities with girls, while the opposite trend was observed for math activities with boys. Additionally, changes in gender differences in children's language and math skills over the period of the study were mediated by language and math parental input, respectively. The discussion explores the implications of these findings for understanding the nature of gender differences and considers potential approaches to addressing these disparities.

The present study investigates phenomena at the intersection of two lines of developmental research, both of which have drawn the attention of scientists and the public alike. The first concerns gender differences in early academic skills and the second concerns a key environmental contributor to early learning, namely, parental input. The public interest in gender differences is understandable, given persistent and well publicized gender gaps, such as higher participation of males in STEM careers. Scientific interest in the topic is largely driven by a desire to understand the environmental and biological roots of individual variability in learning, with parenting arguably the most studied of environmental factors.

By now, a large body of work has documented the role of parents in the growth of early math and language skills (Gunderson & Levine, 2011; Hoff, 2003; Huttenlocher et al., 2002; Lefevre et al., 2009). Yet, there is a paucity of research examining parental input in relation to gender differences in children's academic skills. This is somewhat surprising given that the child's gender is, in fact, an important predictor of variability in family processes, including the level and quality of supports that parent provide to their daughters versus sons (Bornstein, 2013; Endendijk et al., 2016; Marks et al., 2009). In turn, even subtle differences in parents' input could differentially affect children's developmental trajectories

(Mesman & Groeneveld, 2018; Tenenbaum & Leaper, 2002). The current study adds to cumulative knowledge on both gender differences and parenting. It investigates variability in parental input as a function of child gender, and the role of this input as a potential mediator of gender differences in children's skills in key academic domains: math and literacy. By focusing on families in Russia, the study also extends the largely Western-oriented research on parenting and its relation to child outcomes.

1. Gender differences in math and literacy skills: contrasting findings

Much of the literature on gender differences in academic skills is focused on male advantage in mathematics, which emerges early in elementary school (Contini et al., 2017; Fischer & Thierry, 2022) and persists through subsequent years of education (Geary et al., 2023; Wai et al., 2010). Boys have been found to outperform girls on a variety of math tasks and measures, including accuracy and frequency of using advanced problem-solving strategies (Carr & Alexeev, 2011; Casey & Ganley, 2021). It should be noted that there is disagreement regarding the extent of the current gender gap in math achievement, with some

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investigators arguing that it has narrowed or even disappeared (Hyde & Mertz, 2009; Lindberg et al., 2010). Yet, other researchers suggest that the null findings reported in these studies may be due to the nature of assessments, as differences in math performance are more likely to emerge on novel/more challenging problems (Gibbs, 2010; Penner, 2008).

Whereas gender-related variability in math has attracted more attention than gender variability in literacy, the latter has been well-documented in prior research. The pattern observed in the literacy domain is opposite to that in math - a girls' advantage emerges early in language development (Eriksson et al., 2012) and continues across school years (Brozo et al., 2014; Chatterji, 2006). Just like boys' math advantage, girls' advantage in literacy has been reported on a range of tasks, including letter-sound associations, word recognition, writing, and reading comprehension (Ready et al., 2005; Sigmundsson, et al., 2018).

A useful way to juxtapose boys' and girls' achievement in the key academic domains is by examining the results of large-scale international studies, such as TIMMS and PISA (Mullis et al., 2023; OECD, 2018). According to the most recent reports based on the data collected from 4th graders in more than 50 countries, the average math score (across all participating countries) was significantly higher for boys than for girls, whereas the average reading score showed the opposite pattern (Mullis et al., 2020, 2023). In addition to general trends, international studies allow us to examine variability among countries. In math, there were very few countries where girls scored higher than boys; in most countries, including those with distinct socio-cultural contexts (e.g., Russia and the U.S.), the math gender difference favored boys. The results were even more robust in the domain of literacy where no country showed boys' advantage.

Persistent findings of gender differences in academic skills raise questions about their antecedents, and the fact that one major academic domain favors boys while another favors girls makes this question even more interesting. Attempts to identify mechanisms underlying gender variability have ranged from investigating biological predispositions to exploring structural social constraints (Bian, et al., 2017; Geary, 2005; Guiso et al., 2008; Plomin et al., 2016; Vuletich et al., 2020). Recognizing that this complex phenomenon is likely associated with multiple factors at different levels of analysis, we focus in the present paper on a malleable proximal factor - the input children receive during interactions with their caregivers at home. Specifically, we ask whether the activities in which parents engage children at home vary as a function of child gender and, critically, whether this variability in parental input accounts for the observed gender differences in children's skills.

2. Parental input in relation to children's math and literacy skills

In both math and literacy domains, researchers documented a significant relation between parental input and child development during preschool and early school years. In math, investigators identified home activities that involve quantitative components and demonstrated that the frequency of such activities predicts the growth of number skills (Huntsinger et al., 2016; Skwarchuk et al., 2014). Further, the quality and quantity of parental math talk is related to children's understanding of quantitative concepts (Braham et al., 2018; Gibson et al., 2020; Levine et al., 2010). Similarly, the frequency of home literacy activities (e.g., reading or playing word games) is positively associated with children's performance on literacy tasks (Davis-Kean, 2005; Sénéchal & LeFevre, 2014), and the characteristics of parental speech (e.g., lexical diversity and syntactic complexity) predict the growth of children's oral language (Hoff, 2003), which provides the basis for literacy.

The fact that children's skills in both math and literacy domains have been linked to parental input supports the possibility that gender differences in these skills may be driven, in part, by differences in the input parents provide to boys versus girls. This possibility would be further supported by the evidence that parents' investment into home math and

literacy activities varies for boys and girls and that the pattern of this variability is consistent with that observed in children's outcomes. Below, we review findings from existing research on the variability in parental input as a function of child gender.

In considering this evidence, it is important to note, however, that parental input is not the sole environmental factor associated with children's skills; school experiences also play a key role. In most prior research on parental effects (as well as the current study), school input is treated as an omitted variable. Yet, in contrast to the diverse approaches found in U.S. schools, Russian children experience a more standardized curriculum and instruction, both in preschool and elementary grades. This uniformity reduces variability in educational experiences, providing a more controlled context to examine the relation between parental input and children's outcomes.

3. Differences in parental input as a function of child gender

Much of the research on gender-differentiated parenting practices examined psychological differences in parenting styles, such as the amount of parental control, sensitivity, and type of affect (Amicarella et al., 2018; Sikiö et al., 2018). These factors may play a role in children's academic achievement, but it is not clear why their impact would differ across academic domains. Other studies focused on parental expectations of children's success - e.g., higher expectations for boys' math and science achievement (Fryer & Levitt, 2010). Yet, parental expectations may affect child outcomes only indirectly, through specific characteristics of the input children receive via parent-initiated activities or verbal interactions.

A few studies examined parental input as a function of child gender in the context of language/literacy development; the pattern of variability reported in these studies is consistent with children's findings. In particular, mothers of infants initiate more frequent and extended conversations with girls, compared to boys (Clearfield & Nelson, 2006; Sung et al., 2013). Parents of preschoolers are also more likely to engage girls than boys in literacy activities, such as book reading or learning the alphabet (Baker & Milligan, 2016).

With respect to parental math input, few studies have specifically focused on the variability related to child gender; those that have suggest that boys receive more math support at home compared to same-aged girls (e.g., Bhanot & Jovanovic, 2005; Chang et al., 2011). In other investigations of home math environment, where child gender has been included as a control variable, the results are mixed. Some researchers found that boys were exposed to math activities and math talk more than girls (LeFevre et al., 2010; Zippert et al., 2019), while others reported the opposite pattern or no differences (Lu et al., 2023). These inconsistencies may be partly due to differences in methodology and child age; also, many studies involved small samples, potentially lacking the power to detect gender effects.

In sum, the extant findings of parental input variability as a function of child gender are more robust in literacy than in math. However, the fact that most studies reviewed above focused on a single domain (either literacy or math) makes it difficult to compare the patterns of variability across domains. To address this concern, the present study examined gender effects simultaneously (within-participant) in literacy and mathematics - using parallel instruments to assess children's skills and parental input in both domains.

4. Present study

The present study examined gender differences in math and literacy skills in a sample of more than a thousand first-grade students from Russia (data were collected in 2018). According to cross-national studies, elementary school students in Russia demonstrate gender differences in academic achievement, favoring boys in mathematics and girls in literacy (Mullis et al., 2020; Mullis et al., 2023). Investigations of parenting practices suggest that Russian parents engage children in a

variety of activities, similar to those documented in research conducted in Western countries (e.g., joint reading, learning letters, playing board games that involve numbers, etc.); and that the frequency of these activities predict children's academic skills. Given these findings on the relation between home activities and child development, the well-documented gender differences in Russian children's academic skills may be linked to corresponding differences in parental input - this possibility is examined in the present study.

Children were assessed at the start and the end of the school year, which allowed us to examine not only boys' and girls' skills at school entry but also the growth of these skills over time. Based on prior work, we hypothesized that girls would have higher literacy scores and boys would have higher math scores at the first test point, and that these differences would remain, or even increase, at the second time point. To test these hypotheses, we analyzed children's scores focusing on whether (a) the pattern of gender differences varied across domains and (b) the size of gender differences varied between the two time points.

In the middle of the school year, we administered a parental survey that included parallel measures of math and literacy activities. Given that the research on differences in home activities as a function of child gender is limited and the findings in the math domain are inconsistent, we did not have directional hypotheses with respect to parental input. We examined the data to determine whether (a) within each domain, the frequency of home activities varied for boys and girls; and (b) the pattern of gender differences in parental input across the two domains matched that observed in children's skills. To anticipate results, the answers to both questions were positive, allowing us to address our final question.

The key goal of the study was to determine whether gender differences in parental input provide pathways for explaining gender differences in children's literacy and math skills. Having the data in both domains obtained from the same sample made it possible to simultaneously examine the paths from child gender to academic outcomes via parental input in each domain. Further, it allowed us to investigate the possibility of complementarity, for example, the extent to which literacy input may contribute to child math skills. Fig. 1 depicts the conceptual model we used to predict children's scores at the end of first grade. After testing this conceptual model without controlling for children's scores at the start of the academic year, we tested a model that included children's initial scores. This allowed us to determine whether parental input mediated the association between child gender and changes in academic skills above and beyond where children started at the beginning of the academic year.

5. Method

5.1. Participants

The sample included 1,240 first-grade students ($M_{\text{age}} = 7.4$ years,

range = 5.9–8.6, 49.5 % female) and their parents, recruited from 65 classrooms across 29 public schools in a large Russian city. The recruitment was conducted using stratified cluster random sampling: schools were randomly selected from within each of the city's administrative districts. Location-based stratification was intended to yield a socio-economically diverse sample. Indeed, the resulting sample represented a broad range of economic backgrounds: 25 % low-SES, 45 % middle-SES, and 30 % high-SES level (based on household income reported by parents). Although individual information about participants' race/ethnicity was not collected, the city population is largely homogeneous in terms of ethnicity and native language (91 % Russian).

5.2. Measures

5.2.1. Child assessment

First-graders' skills were assessed twice - at the start and at the end of the school year (Time1 and Time 2, respectively) - using a Russian version of the International Performance Indicators in Primary Schools, which showed high predictive validity and internal consistency (0.92) in previous work (Tymms et al., 2015). The test instrument included a literacy section covering letter identification, word recognition and reading comprehension; and a math section covering symbolic number knowledge (number recognition and counting), basic arithmetic (addition and subtraction), and mathematical reasoning (arithmetic word problems). Testing was conducted in schools and lasted approximately 30 min per child. The rules of test administration were the same for both sections. At Time 1, participants started with the first item within a section; they were stopped after making three errors in a row. At Time 2, the start item varied, depending on where the participant stopped at Time 1. For each section, a raw score was computed as a total number of correct responses and then transformed into a standardized score on an equal-interval scale using the dichotomous Rasch model (Fischer & Molenaar, 2012). Thus, each child had four scores - one for each domain (literacy and math) at each testing point.

5.2.2. Parent survey

In the middle of the school year, parents received a questionnaire in two parts. The first part contained questions about the family background and literacy-related activities. The second part, completed within a month of the first one, contained questions about math-related activities.

With respect to family background, the variable examined in the present study was the level of education of the primary caregiver. Parents were asked to indicate their level of education on a 7-point scale; their responses were coded as the number of years corresponding to the reported level of education: 9 = incomplete high school (minimal required education), 11 = high school diploma, 13 = vocational education, 14 = at least 3 years of college, 15 = college degree, 17 = Masters or Professional degree (e.g., medical doctor), 20 years = Ph.D.

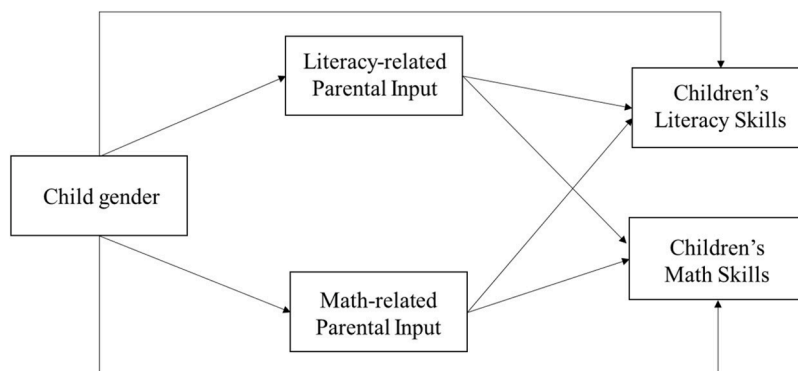


Fig. 1. Conceptual model.

The home activity questionnaires presented parents with a list of activities, nine of which centered on literacy and nine on math (see Supplementary Materials for the items). The list was preceded by the question: “In the last six months, how often did you, or another member of your family, engage in the following activities with your child?” Parents indicated the frequency of each activity on a 7-point scale: 1 = never; 2 = very rarely; 3 = 1–3 times per month; 4 = once a week; 5 = 2–3 times per week; 6 = once a day; 7 = more than once a day. Both literacy and math scales showed good internal consistency with Cronbach’s alphas of 0.85 and 0.88, respectively. For each parent, two scores were computed - one representing the average frequency reported on the nine literacy items and the other representing a parallel score for the math items.

6. Results

6.1. Gender differences in children’s academic skills

Table 1 presents descriptive statistics along with *p*-values and effect sizes for gender differences in the key predictors and outcome variables. As could be expected, scores in both literacy and math increased over the school year. Moreover, consistent with our hypotheses, gender differences showed the opposite pattern across domains, with girls demonstrating higher scores in literacy and boys in math (see Table S1 in the Supplementary Materials for zero-order correlations among the variables listed in Table 1).

As a follow-up to examining these gender differences, scores were submitted to a 2 (Gender) x 2 (Domain) x 2 (Time) ANCOVA with child’s age at T1 and parent’s education as covariates. With respect to covariates, child’s age was not associated with the scores, whereas parental education was, $F(1, 1236) = 128.13, p < .001, d = 0.32$ - parents with higher levels of education had children with higher scores. With respect to the key predictors, time was the only significant main effect, $F(1, 1236) = 4623.79, p < .001, d = 1.94$. Gender and domain, while not having main effects, produced a significant interaction, $F(1, 1236) = 127.03, p < .001, d = 0.32$. Follow-up simple-effects tests showed that the gender difference was significant within each domain, with girls demonstrating an advantage in literacy and boys in math.

In addition to the two-way interaction, the three-way interaction of gender x domain x time was also statistically significant, $F(1, 1236) = 16.07, p < .001, d = 0.16$, suggesting that the size of gender differences changed between the two time points. Indeed, follow-up analyses within each domain showed a gender x time interaction both for literacy $F(1, 1236) = 5.63, p = .02, d = 0.07$; and math $F(1, 1236) = 7.97, p = .005, d = 0.08$. Gender differences were significant in each domain at T1, but the magnitude of these respective disparities in literacy and math

increased further across the year (see Fig. 2).

6.2. Differences in parental input as a function of child gender

Following the analysis of children’s scores, we tested the hypothesis that parental input would vary by gender such that girls would receive more verbal input and boys would receive more quantitative input. Consistent with this hypothesis, as shown in Table 1, literacy-related parental input was higher for girls, whereas math-related input was higher for boys. Submitting parental input to a 2 (Domain) x 2 (Child Gender) ANCOVA with parent education and child age as covariates, we found a main effect of domain, $F(1, 1237) = 88.15, p < .001, d = 0.27$, indicating that, overall, parents produced more input in the literacy domain, compared to math.

Further, there was a significant interaction between domain and gender, $F(1, 1237) = 130.49, p < .001, d = 0.33$, indicating that child gender was differentially associated with verbal versus quantitative input. Simple-effects tests showed that parents produced more literacy input for girls than for boys, and vice versa for math input (both *p*’s < 0.01). In addition, with respect to the covariates, parent’s education was significantly associated with parental input, $F(1, 1237) = 60.87, p < .001, d = 0.22$, whereas child’s age was not, $F(1, 1237) = 2.076, p = .15, d = 0.05$.

6.3. Relation between gender differences in parental input and children’s academic skills

To address the key research questions, we estimated maximum likelihood path models, examining the direct and indirect associations between child gender, literacy- and math-related parental input, and child’s literacy and math skills (see Fig. 1 for the hypothesized model). We included child age and parent education as covariates for all predictor, mediator, and outcome variables. Bias-corrected bootstrapping (10,000 draws) was used to generate confidence intervals for direct and indirect effects (Dearing & Hamilton, 2006; Hayes et al., 2011). All analyses were performed using Mplus 8.3s (Muthén and Muthén, 2019).

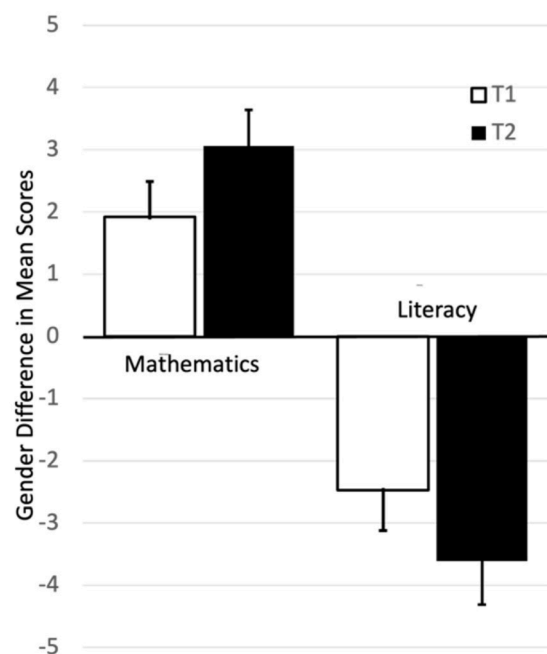


Fig. 2. Gender differences in children’s performance in two domains at two time points.

Note. Each bar depicts the difference in the mean level of performance between boys and girls, computed as: Mean (boys) - Mean (girls).

Table 1

Descriptive statistics, significance levels (*p*) and effect sizes (*d*) for gender differences.

Domain (Time)	Overall M (SD)	Girls M (SD)	Boys M (SD)	<i>p</i>	<i>d</i>
Children’s Scores					
Literacy (T1)	50.33 (10.25)	51.58 (10.31)	49.11 (10.04)	<0.001	0.24
Literacy (T2)	62.14 (12.46)	63.96 (12.65)	60.35 (12.01)	<0.001	0.29
Mathematics (T1)	50.11 (10.04)	49.14 (9.16)	51.05 (10.75)	.001	0.19
Mathematics (T2)	61.54 (11.45)	60.00 (10.41)	63.06 (12.21)	<0.001	0.27
Parental Input					
Literacy-related	51.26 (9.97)	52.64 (9.51)	49.91 (10.24)	<0.001	0.28
Math-related	48.84 (10.19)	47.21 (10.33)	50.44 (9.80)	<0.001	0.32

Standardized coefficients are presented in Fig. 3. Consistent with ANOVA results, path analyses showed a positive direct association between child gender and academic skills. In particular, child gender predicted literacy scores showing a girls' advantage and math scores showing a boys' advantage. Similarly, child gender was directly associated with literacy input favoring girls and with math input favoring boys. Regarding the cross-paths between parental input and children's performance, there was a positive association between literacy input and child mathematics skills, whereas there was no relation between math input and literacy skills.

With regard to our hypothesis that parental input would mediate links between gender and children's scores in literacy and math, the indirect effect from child gender to literacy skills was mediated by literacy-related parental input but not via math-related parental input. In addition, the indirect effect from child gender to math skills was significantly mediated by both literacy-related and math-related parental input, and the total indirect effect of these two paths was also statistically significant (see Table 2).

As a final step of analyses, we respecified our path model to include child scores at Time 1 as covariates. That is, we estimated the indirect path: child gender → parental input → child skills, while including Time 1 scores as predictors of Time 2 scores (Fig. 4). By doing so, our model estimated residual change in literacy and math, thereby allowing us to examine whether changes in gender differences in child skills over the year were mediated by parental input. In addition, because we also included a path from child scores at Time 1 to parental input, these respecified models provided a rigorous control for omitted variables, ruling any factors affecting children's initial skill levels and evocative effects on parent input (NICHD ECCRN & Duncan, 2003). Note, however, that for reasons of parsimony we removed the cross-path between literacy-related parental input and math skills in Time 2 that was not significant in our previous (or this) model. Standardized coefficients of the direct pathways are presented in Fig. 4.

As shown in Fig. 4, when children's scores at Time 1 were included in the model, the direct effects between child gender, parental input, and children's scores all remained statistically significant. However, the coefficients of the direct effects between parental input and children's skills dropped in magnitude. A similar finding was obtained with regard to the indirect effects. That is, both indirect pathways between child gender and changes in academic skills (math and literacy) via corresponding measures of parental input were statistically significant, yet

Table 2

Path analysis results without controlling for children's skills in T1.

Indirect paths			95 % CI	
	Estimate	SE	Lower	Upper
<i>From child gender to literacy skills</i>				
Child gender → Literacy-related parental input → Literacy skills	0.027	0.007	0.016	0.040
Child gender → Math-related parental input → Literacy skills	0.003	0.006	-0.006	0.014
TOTAL:	0.144	0.023	0.107	0.183
TOTAL indirect:	0.030	0.010	0.014	0.048
<i>From child gender to math skills</i>				
Child gender → Literacy-related parental input → Math skills	0.008	0.005	0.001	0.016
Child gender → Math-related parental input → Math skills	-0.015	0.006	-0.025	-0.005
TOTAL:	-0.137	0.026	-0.180	-0.095
TOTAL indirect:	-0.006	0.009	-0.022	0.009

with smaller effect sizes than prior to controlling for Time 1 skills (see Table 3).

7. Discussion

The goals of this study were two-fold. One was to establish the magnitude and direction of gender differences in children's math and literacy skills, as well as in parental input provided in the corresponding domains. These issues have been addressed in prior research, but the studies typically focused on one domain at a time and the findings, especially in the domain of math, were not always consistent. The other goal was to test parental input as a mediating pathway between child gender and academic skills. Investigating this issue in two domains that were expected to show the opposite pattern of gender differences was the main novel contribution of the study.

7.1. Variability in children's academic skills as a function of gender

First-grade girls in our study had higher scores on the literacy test but lower scores on the math test compared to same-age boys. In the domain of literacy, the results are consistent with prior findings obtained across different ages, measures, and even across countries. In the domain of math, while many studies reveal the same pattern, others have not (e.g.,

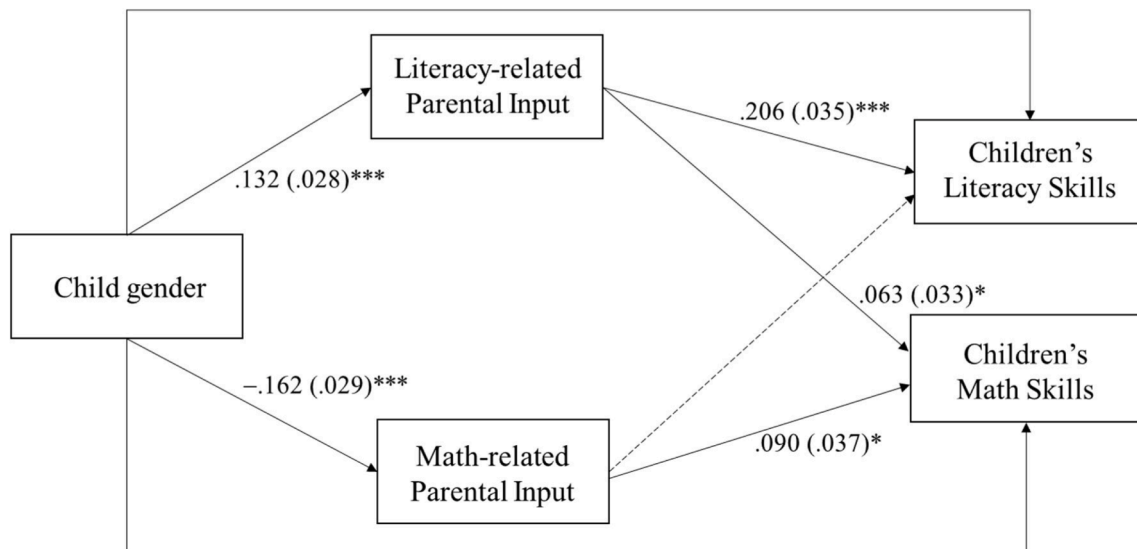


Fig. 3. Path analysis results (not controlling for children's initial skills).

Note. Standard error estimates, in parenthesis, were adjusted by classroom clusters. The model included covariance between outcomes and between mediators, and error correlations between domains (not shown in the Fig.). Non-significant paths are shown using dotted lines. * $p < .05$, ** $p < .01$, *** $p < .001$.

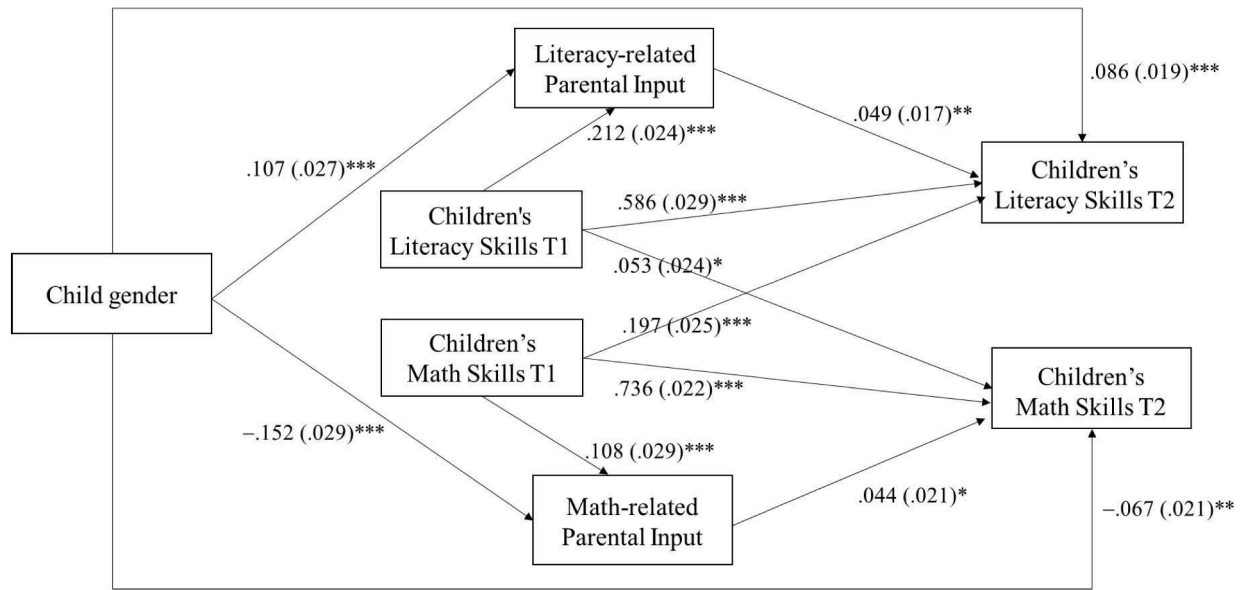


Fig. 4. Path analysis results (controlling for children's initial skills).

Note. Standard error estimates, in parenthesis, were adjusted by classroom clusters. The model included covariance between outcomes, covariance between mediators, and error correlations between domains (not shown in the figure). * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 3

Path analysis results controlling for children's skills in T1.

Indirect paths			95 % CI	
	Estimate	SE	Lower	Upper
<i>From child gender to literacy skills</i>				
Child gender → Literacy-related parental input → Literacy skills	0.005	0.002	0.002	0.009
TOTAL	0.091	0.019	0.061	0.123
TOTAL indirect	0.005	0.002	0.002	0.009
<i>From child gender to math skills</i>				
Child gender → Math-related parental input → math skills	-0.007	0.004	-0.013	-0.001
TOTAL	-0.073	0.020	-0.106	-0.039
TOTAL indirect	-0.007	0.004	-0.013	-0.001

Hyde & Mertz, 2009). It has been proposed that the inconsistency in math gender findings may be driven, in part, by the level of test difficulty (e.g., Ganley & Vasilyeva, 2014). The test utilized in the current study included items that varied in difficulty from moderately to highly challenging; in fact, no participant solved all the items correctly. This, as well as having a large sample size, may have allowed us to capture the difference between boys' and girls' math performance in first grade.

The size of gender differences obtained in the present study was relatively small (with Cohen's d varying between 0.17 and 0.29), as it tends to be in other investigations of gender effects on academic skills (e.g., Ardila et al., 2011; Peterson, 2018). Relatively small effect sizes may prompt questions about the importance of studying this phenomenon. Yet, small gender differences in early academic skills may translate into increased anxiety and decreased interest, which may in turn impact later achievement and career choice. Indeed, prior research has established a bidirectional link, whereby poor math performance early on leads to math anxiety, which hinders math learning (Ma & Xu, 2004; Zhang et al., 2019). Further, growing evidence suggests that gender gaps in STEM participation may have roots in early math self-concepts, which affect interests and academic choices throughout school and into college (e.g., Cvencek et al., 2021; Wang & Degol, 2013). Given these cascading consequences, it is understandable why researchers' attention has turned to how early gender differences in academic skills arise. Gender-differentiated input in the early home environment offers one

potential mechanism.

7.2. Variability in parental input as a function of child gender

Overall, parents in the present study engaged children in literacy activities more often than in math activities. Within each domain, the pattern of gender differences mirrored children's performance: girls, on average, received more literacy than math input, whereas boys received more math than literacy input. What makes these findings particularly interesting is that the information about both domains was obtained from the same group of parents. This minimizes the possibility that one set of parents (e.g., parents of girls) were overall more likely to engage children in academic activities than the other set of parents. It also minimizes the possibility that more general parenting characteristics, such as parental affect or amount of control, contributed to gender differences in children's outcomes because presumably these factors should have a comparable effect on the development of children's math and literacy skills.

On the other hand, cultural stereotypes and expectations may play a role in helping explain how and why gender differentiated parenting arises (Bornstein, 2013; Vuletic, et al., 2020). In Russian culture, there is a tension between the stated goals of gender equity and a persistent endorsement of the traditional stereotypes of masculinity vs. femininity (Kerig et al,1993). For example, it is considered more acceptable to discuss emotions and interpersonal relations with girls than with boys. This tendency, which has been also documented in other cultures (Aznar & Tenenbaum, 2015), may certainly contribute to a greater amount of verbal interaction and higher frequency of story-telling and conversations about past events with girls.

Yet, another plausible explanation of differences in parenting as a function of child gender has to do with child elicitation, or evocative, effects. In the literacy domain, girls may elicit a greater amount of language input as they tend to start speaking at an earlier age (Johnson et al., 2014). Child elicitation effects, however, do not appear to fully explain gender differentiated parenting in this domain: parents have been shown to initiate more frequent and longer "conversations" with girls than with boys even when the children are at preverbal stages of development (e.g., Clearfield & Nelson, 2006; Johnson et al., 2014). With respect to the quantitative domain, there is little evidence of gender differences in very young children, but there is evidence of boys'

early advantage in spatial skills (Lauer et al., 2019; Levine et al., 1999). Given a large body of research demonstrating a link between spatial and quantitative reasoning (Mix & Cheng, 2012), more advanced spatial skill may confer advantages on boys' quantitative skills, which in turn, may encourage parents to engage them in math activities.

Generally, the examination of child-elicitation effect is complicated by the bi-directional nature of the relation between parental input and child development. Beyond early infancy, differences in child skills are likely driven, at least in part, by parenting, while the differences in parenting are driven, in part, by child's own behavior (Bornstein & Putnick, 2016). While not a panacea for disentangling this complexity, multiple measurements of child skills can help, particularly if these measurements are captured prior to and after parenting practices have been assessed. Below, we present our results based on using such an approach.

7.3. Parental input as a pathway for gender differences in children's academic skills

To examine the extent to which the frequency of home activities can account for gender differences in math and literacy skills, we ran two sets of models. The first set, which did not control for children's initial skills, produced clear evidence of significant indirect effects in both literacy and math domains. This finding is consistent with the possibility that gender differences in children's skills at the end of first grade are driven, in part, by the variability in parental input. A plausible concern about this interpretation is that the variability in parental input itself may be driven by children's prior skills, as discussed above with regard to child-evocative effects.

To account for these evocative effects, our second set of models included child scores at the start of first grade as controls for both parental input and children's scores at the end of the first grade. Thus, in this second model, we examined the extent to which parental input predicted changes in gender differences in math and literacy skills over the academic year, while holding constant the role of child initial skills in evoking parental input. As could be expected, children's initial skill levels in each domain significantly predicted their later skills, as well as parental input, in the corresponding domain. Adding this strong control variable decreased the indirect effects between child gender, parental input, and academic skills, and made the cross-domain indirect effect from child gender to math skills via literacy input no longer significant. Still, in both domains, the domain-specific indirect paths from child gender to academic skills via parental input remained significant.

In considering the findings from both sets of analysis, it is important to acknowledge that while controlling for the initial skills helps us rule out child evocative effects (as well as other omitted variables that influence initial skill levels), doing so also makes the model very conservative. Differences in children's skills at the start of the first grade may themselves be driven by prior parental input. Thus, when controlling for initial skills, we may be eliminating the variability in children's skills at the end of first grade that is due to accumulating parental input from prior years. The question of which approach to choose is a recurring dilemma in the developmental research. Arguably, by using both approaches in the present study, we have estimated what are perhaps upper and lower bounds for the true effect of parental input. The model not controlling for evocative effects is likely an upper bound of the parental effect, and the model controlling for these effects likely provides a lower bound. With this mind, if the true effects fall somewhere in between these bounds, the estimated effect sizes for parental input appear meaningful for both theoretical understanding and practice.

Regarding effect size, it is also important to consider what we have captured with our parental input measure. While measurement error is a concern in all psychological studies, assessing the full contribution of parenting to children's development is particularly challenging. Like most studies of parenting, we are only able to capture a "snapshot" of parenting behaviors experienced by the child. Unquestionably, these

empirical snapshots exist within a large parenting milieu in which parents socialize and educate their children through a rich portfolio of experiences, interactions, conversations, behaviors, and structuring of the environment (Dearing & Tang, 2010). It is reasonable to assume that the total effect of the full portfolio of inputs parents provide is stronger than that of any snapshot estimates.

It is instructive to consider the present findings in the context of cross-national research showing that while there are clear trends in the patterns of gender differences in academic achievement (consistent with those observed in the present study), there is also substantial variability among countries in terms of effect sizes (Mullis et al., 2020; Rey-Guerra et al., 2023). This variability is often discussed in terms of macro-level socio-cultural differences, such as gender-equality index. Yet, to better understand the psychological mechanisms underlying a particular pattern of gender differences emerging in childhood, it is important to look at processes that are more proximal and, potentially, more malleable. Parental input is one of the key proximal factors in child development and there is accumulating body of research suggesting ways in which it can be shaped to improve children's outcome through training or even through designing play materials with specific features (Laski & Siegler, 2014; Lu et al., 2023; Vasilyeva et al., 2006). The present findings bring up questions about the extent of malleability of the particular aspects of parental input (e.g., frequency of specific activities or characteristics of parental talk) that vary as a function of child gender. Addressing these questions is an important step that may have far-reaching consequences for children's success in school and beyond.

8. Limitations and future directions

The mediating pathways explored in this study represent just one aspect of the complex puzzle of gender differences in key academic domains. There are dimensions of parental input not captured by our measures, such as the quantity and quality of math talk or differences in toy selection for sons versus daughters (MacPhee & Prendergast, 2019). Additionally, contextual factors like teachers' pedagogical approaches and cultural norms conveyed through media likely contribute to gender differences in math and literacy (Brown & Stone, 2018). Ideally, future research should investigate multiple potential mechanisms underlying these differences. It is also plausible that the role of contextual factors varies across time, with parental effects being more pronounced early on and media effects gaining prominence later in the development. Testing this possibility would require a longitudinal approach, which was beyond the scope of the current study. Expanding to multiple waves of data collection would further strengthen this type of work by clarifying, for example, the extent to which parental input is evoked by children.

It should be also noted that even though structural equation modeling, which was used to analyze indirect effects in the present study, has important strengths, we cannot rule out unobserved time-varying potential confounds, and therefore, variability in parental input and children's outcomes may have been driven, in part, by unobserved time-varying factors. Still, this approach provides an important step, which can be complemented by more focused experimental studies targeting specific aspects of parental input. To identify potentially malleable factors associated with parental input, it may be particularly useful to examine the relation between cultural beliefs (in particular, gender stereotypes in different domains) and gender differences in parental input and child outcomes, in these domains. Cross-cultural research may provide opportunity for this type of investigation.

9. Conclusion

Gender differences in early academic skills remain an important concern for researchers and educators alike (e.g., Kroeger et al., 2019). Indeed, there is increasing evidence that early differences in math and literacy can have cascading consequences for long-term achievement and well-being (e.g., Cvencek et al., 2021; Gibb et al., 2008). Given that

literacy differences favoring girls and math differences favoring boys emerge early in life, it is critical that we understand whether and how early parenting behaviors may help establish these gender differences. In the present study, we find robust evidence that girls' and boys' academic skills do indeed differ at first grade in the expected directions, and that gender differentiated parenting accounts for a significant portion of this gender-related variability. Consistent with a large literature on the importance of the early home environment for children's achievement, our findings underscore the potential value of intervention efforts aimed at the home environment.

CRedit authorship contribution statement

Marina Vasilyeva: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Catalina Rey-Guerra:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Linxi Lu:** Writing – review & editing, Writing – original draft, Formal analysis. **Eric Dearing:** Writing – review & editing, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecresq.2025.02.002](https://doi.org/10.1016/j.ecresq.2025.02.002).

Data availability

Data will be made available on request.

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